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PostgreSQL Linux Setup and Basics: From Zero to Ready — PostgreSQL Foundations Series

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Published Oct 24, 2025

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PostgreSQL is a powerful, open-source relational database trusted by developers, analysts, and enterprises worldwide. When paired with Linux, it becomes a secure, flexible, and performant data platform that can scale from a Raspberry Pi to massive cloud clusters.

This guide walks you through how to **install, configure, and start using PostgreSQL** on

🧩 1. Installing PostgreSQL on Linux

PostgreSQL is available in the default repositories of most major Linux distributions.

🟠 Ubuntu / Debian

```
sudo apt update.
sudo apt install postgresql postgresql-contrib
```

This installs the latest PostgreSQL version available in the distro's repo. To get newer versions:

```
# Add PostgreSQL APT repository
sudo sh -c 'echo "deb http://apt.postgresql.org/pub/repos/apt $(lsb_release

wget --quiet -O - https://www.postgresql.org/media/keys/ACCC4CF8.asc | sud
sudo apt update
sudo apt install postgresql-16
```

🔵 Red Hat / CentOS / Fedora

```
sudo dnf install postgresql-server postgresql-contrib
sudo postgresql-setup --initdb
sudo systemctl enable postgresql
sudo systemctl start postgresql
```

👤 2. Working with the PostgreSQL User

After installation, a Linux user named postgres is created. Switch to it:

```
sudo -i -u postgres
```

To open the PostgreSQL CLI:

```
psql
```

To exit:

```
\q
```

Create a new database and user:

```
CREATE DATABASE mydb;
CREATE USER wayne WITH ENCRYPTED PASSWORD 'securepassword';
GRANT ALL PRIVILEGES ON DATABASE mydb TO wayne;
```

⚙️ 3. Basic Configuration (postgresql.conf, pg_hba.conf)

PostgreSQL's configuration files typically reside in:

- /etc/postgresql/<version>/main/ (Debian-based)
- /var/lib/pgsql/data/ (Red Hat-based)

Key files:

- postgresql.conf: General config (ports, memory, logging)
- pg_hba.conf: Client authentication rules

🔧 **Example changes:**

Enable remote access:

```
# postgresql.conf
listen_addresses = '*'
```

Allow IPs to connect:

```
# pg_hba.conf
host    all             all             192.168.1.0/24          md5
```

Reload config:

```
sudo systemctl reload postgresql
```

🧱 4. Basic SQL Commands

Create a table:

```
CREATE TABLE users (
  id SERIAL PRIMARY KEY,
  name TEXT NOT NULL,
  email TEXT UNIQUE NOT NULL,
  created_at TIMESTAMP DEFAULT now()
);
```

Insert data:

```
INSERT INTO users (name, email) VALUES ('Alice', 'alice@example.com');
```

Query:

```
SELECT * FROM users;
```

Index:

```
CREATE INDEX idx_users_email ON users(email);
```

📊 5. Useful Linux Commands for PostgreSQL Admins

```
# Check status
sudo systemctl status postgresql

# Check current connections
sudo -u postgres psql -c "SELECT * FROM pg_stat_activity;"

# Check running version
psql --version
```

💾 6. Backups and Restore

Dump a database:

```
pg_dump -U postgres -F c mydb > mydb.backup
```

Restore:

```
pg_restore -U postgres -d mydb_new mydb.backup
```

🔄 7. Start/Stop/PostgreSQL on Linux

```
sudo systemctl start postgresql
sudo systemctl stop postgresql
sudo systemctl restart postgresql
sudo systemctl enable postgresql
```

🧩 8. PostgreSQL Package Ecosystem

Install useful extensions:

```
sudo apt install postgresql-contrib
```

Enable them:

```
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "pg_stat_statements";
```

🗂️ 9. PostgreSQL Folder Layout on Linux

🔧 10. Troubleshooting Tips

- Use pg_isready to check service health
- Tail logs: tail -f /var/log/postgresql/postgresql-*.log
- Clear connections: pg_terminate_backend()

✅ Summary

With these basics, you're ready to explore PostgreSQL in production or development.

#postgresql

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